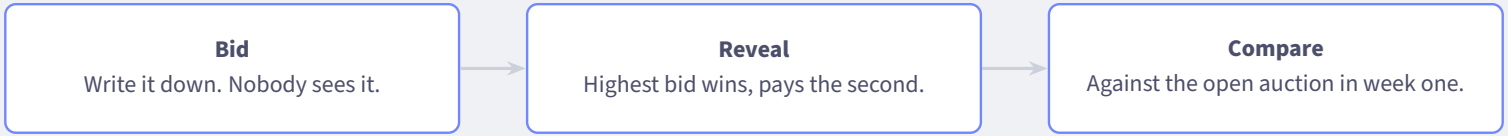
Auctions

The same five pounds, sold a different way



On sale: a five pound note. What is it worth to you?

The bids

Bidder	Bid	Bidder	Bid

Winner | Highest bid | Price paid | Their profit

While you play

Write one bid on a piece of paper and keep it hidden. The highest bid wins the $5 and pays the second-highest bid. Nothing is revealed until every bid is in.

Everyone submits _____, without seeing anyone else's.

The item goes to _____.

The winner pays _____.

In week one we ran this open and ascending, and it went badly. Name one thing that changed when we sealed it:

Your value is v. Let b be the highest bid among everybody else. Compare bidding v with bidding something else.

You bid above your value

You only win extra times when b is between ____ and ____, and then you pay ____ and make ____.

You bid below your value

You lose some auctions you would have won, and when you do still win you pay ____.

So bidding your true value is _____.

If truth is a weakly dominant strategy, then to bid well you need to know _____ about the other bidders.

So the seller gets the item to the person who _____, which is not guaranteed otherwise.

Now change one rule: the winner pays their *own* bid. Is truth still weakly dominant? What should you do instead?

What we just did in the room is **Question 1** on the Auctions page, with values $v_1 = 10$, $v_2 = 7$ and $v_3 = 4$.

(a)	The rules of a second-price (Vickrey) auction.	[3]
(b)	With truthful bids: the winner, the price and the payoff.	[3]
(c)	Show bidding $v_1 = 10$ is weakly dominant, above and below.	[7]
(d)	Why that is a desirable property.	[4]
(e)	Would truth still be weakly dominant in a first-price auction?	[2]

vknight.org/gt/topics/auctions.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Collecting the sealed bids	**16.2.1**	Auction Games: Definition: Auction Game
Bidding your own value	**16.2.3**	Auction Games: Theorem: Bayesian Nash Equilibrium for Second Pay Auction
What you need to know about others	**16.2.2**	Auction Games: Definition: Bayesian Nash Equilibrium
Bidding below value in a first-price sale	**16.2.4**	Auction Games: Theorem: Bayesian NE in a first-price auction
Examinable	**16.2.3, 16.2.4**	Both with proof, and 16.2.5 Revenue Equivalence, statement only

vknight.org/gtb/chapters/auction-games

Quick quiz before you go: vknight.org/gt/topics/auctions.html

Before the next class: read the Auction Games chapter as far as revenue equivalence, and attempt Question 1.